

Broadway Lotteries

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There are k identical tickets to be allocated. There are M individuals. Individual i requests n_i tickets. Let $N = \max_i n_i$.

1. Each individual can apply once, and can list the number of tickets requested. For each individual i , generate $l_i \sim \text{Exp}(1)$, and rank individuals
2. Each individual can apply once. Draw lotteries from $U[0, 1]$ and scale appropriately.
3. Each individual can apply once. Give $N!/n_i$ tickets.
4. Each group can apply once. Hold uniform random lottery.

Assume that we draw until number of remaining seats is less than N (then can discuss non-wasteful version).

Theorem 1 *Methods 1, 3, 4 are equivalent. (But 4 is a pain to verify)*

Not quite – under 4, groups can win multiple times.

Theorem 2 *For any profile of requests, let p_i be the probability of winning for agent i . Fix k . Under exponential draws, there exists c (depends on k , or N ?) such that $|p_i/p_j - n_j/n_i| \leq c/M$.*

Theorem 3 *For any profile of requests, let p_i be the probability of winning for agent i . Suppose that $k < \sum_i 1/n_i$.*

Under uniform draws, there exists c such that $|p_i/p_j - n_j/n_i| \leq c/M$. Key here is that we allow $k = \mathcal{O}(1)$ or k growing with M – linearly or otherwise.

Theorem 4 *Under uniform draws, with k constant and M growing, things look like exponential. Could potentially get Theorem 2 as consequence of Theorems 3 and 4.*

There are m groups of people who want to go to the show together. Group j consists of n_j individuals. Let $N = \max_i n_i$. Consider the following proposals: